NEERAJ SINGH
DEVOPS

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YOUR DATABASE HAS 100 MILLION ROWS!

Would you copy ALL of them every 5 minutes? !?

NO! ✗



Production Database
100 Million Rows

Only a few records change...

- **INSERT** (New data)
- **UPDATE** (Modified data)
- **DELETE** (Removed data)

CDC

Captures only the changes!



Data Warehouse
- Analytics
- Reporting
- Business Intelligence



Data Lake
- Store raw data
- Historical data
- Advanced analytics



Real-time Systems
- Applications
- Dashboards
- Event-driven architecture

CHANGE DATA CAPTURE (CDC)

MOVE ONLY WHAT CHANGED!

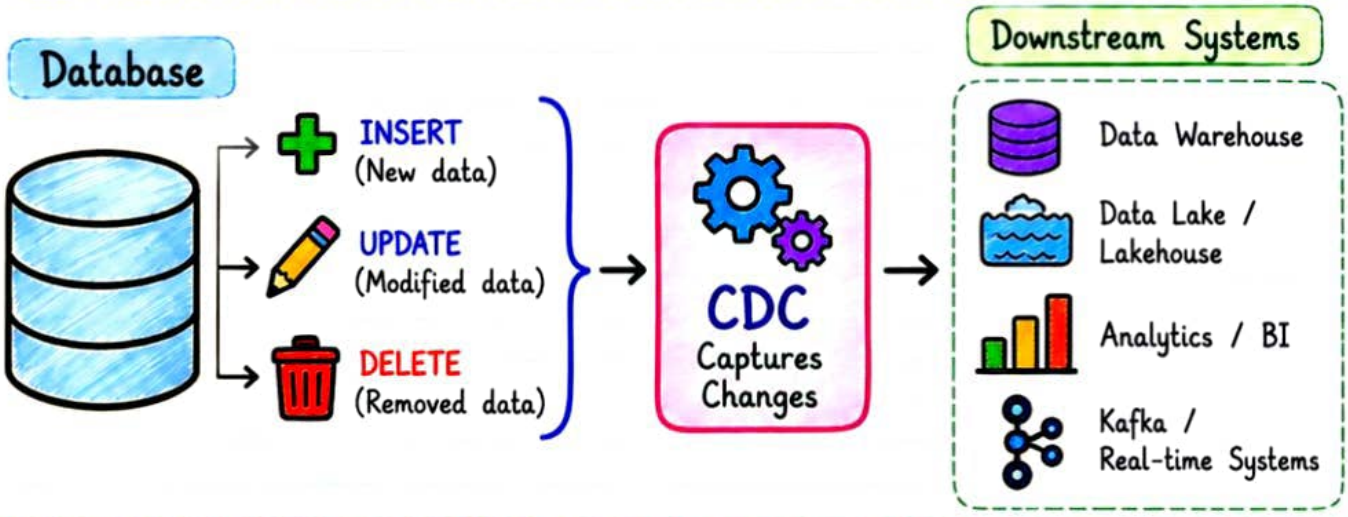
Smarter Data Movement
Bigger Possibilities!

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WHAT IS CDC?

CDC = Change Data Capture

A technique for identifying and capturing changes made to a database and sending only those changes to downstream systems in near real time.



Why do we need CDC?

- ✓ Avoid copying the entire dataset
- ✓ Reduce data movement and storage cost
- ✓ Get near real-time data
- ✓ Keep downstream systems in sync
- ✓ Enables event-driven architectures
- ✓ Supports scalable and efficient pipelines

Common Use Cases

- E-commerce (orders, customers, inventory)
- Banking (transactions)
- CRM (customer updates)
- IoT (sensor data changes)
- Any application where data is frequently changing

Instead of moving everything, CDC captures only the changes!

"Same data. Bigger possibilities. With CDC!"

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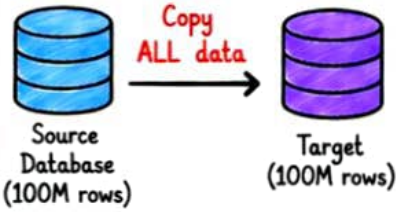
HOW SHOULD WE MOVE DATA?

Full Load vs Incremental Load vs CDC

1. FULL LOAD

Move EVERYTHING

- Copies the entire dataset from source to target every time.
- Used when data is small or for initial load.

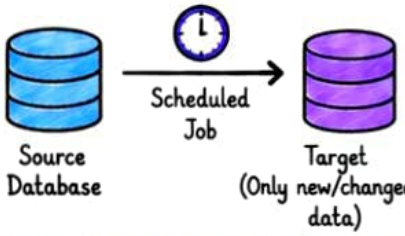


- ✓ Simple to implement
- ✓ Useful for initial load
- ✗ High time and cost
- ✗ Not practical for large data
- ✗ Increases compute and storage usage

2. INCREMENTAL LOAD

Move only selected changes periodically

- Copies new or modified records since the last run.
- Usually runs on a schedule (e.g., hourly, daily).

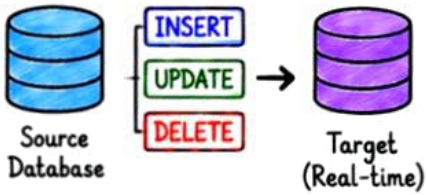


- ✓ Moves less data
- ✓ Reduces cost and time
- ✓ Good for batch pipelines
- ✗ May miss real-time changes
- ✗ Needs logic to identify changes (e.g., timestamp, flags)

3. CHANGE DATA CAPTURE (CDC)

Capture changes continuously

- Identifies and captures INSERT, UPDATE, DELETE events as they happen.
- Typically uses database transaction logs (log-based CDC).



- ✓ Near real-time data movement
- ✓ Captures all changes (I/U/D)
- ✓ Efficient and scalable
- ✓ No need to query the source
- ✗ More complex to set up
- ✗ Requires log access and specialized tools (e.g., Debezium)

QUICK COMPARISON

Feature	Full Load	Incremental Load	CDC
Data Moved	Everything	Selected changes	Only changes
When to Use	Initial load / Small data	Periodic updates (batch)	Real-time (continuous)
Detects Changes	No	Yes (e.g., timestamps)	Yes (log-based)
Latency	High	Medium	Low (near real-time)
Complexity	Low	Medium	High

FULL = EVERYTHING | INCREMENTAL = CHANGES PERIODICALLY | CDC = REAL-TIME CHANGES

Choose the right approach for your use case!

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CDC ARCHITECTURE

How Change Data Capture Works?

Capture database changes → Read logs → Publish events → Send to target systems

1. Source Database

- MySQL
- PostgreSQL
- Oracle
- SQL Server

etc.

Application makes changes in database

2. Database Transaction Log

Transaction Log

INSERT ...
UPDATE ...
DELETE ...

Changes are written to the database logs (e.g., binlog, WAL, redo log)

3. CDC Connector

- Reads the transaction logs
- Identifies the changes
- Converts them into events (messages)
- Publishes to Kafka or target system

Tools: Debezium
Kafka Connect
AWS DMS, etc.

4. Apache Kafka (Streaming Layer)

customer-events
order-events
inventory-events
payment-events

- Stores change events in topics (partitioned)
- Allows multiple consumers to read the events

5. Downstream Systems

- Data Warehouse (e.g., Snowflake, Redshift, BigQuery)
- Data Lake (e.g., S3, ADLS)
- Analytics / BI (e.g., Power BI, Tableau)
- Real-time Applications
- ML / AI Pipelines

Types of CDC

1. **Log-based CDC**
   - Reads database transaction logs
   - Most common and efficient
2. **Trigger-based CDC**
   - Uses database triggers
   - Simple but can impact performance
3. **Query-based CDC**
   - Periodically queries for changes
   - Easier to set up but less efficient

Key Components

1. **Source Database** – where changes happen
2. **Transaction Log** – records all changes
3. **CDC Connector** – reads logs and creates events
4. **Kafka** – stores and streams events
5. **Consumers** – read from Kafka
6. **Target Systems** – warehouse, lake, analytics, etc.

Why Kafka for CDC?

- ✓ High throughput
- ✓ Scalable
- ✓ Reliable and durable
- ✓ Supports multiple consumers
- ✓ Decouples source and target systems
- ✓ Real-time data movement
- ✓ Perfect for event-driven architectures

Database Changes → Events → Real-Time Possibilities!

"It's not just data movement. It's real-time intelligence."

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REAL-WORLD EXAMPLE
E-COMMERCE ORDER FLOW WITH CDC

See how only the changes are captured and sent to downstream systems

1 Customer places an order



→ Production Database

orders	
id	5001
customer_id	101
amount	2500
status	CREATED
created_at	2024-01-15

→ CDC detects INSERT event

New record added to orders table

2 Order status is updated

Order is packed and shipped

→ Production Database

orders	
id	5001
customer_id	101
amount	2500
status	SHIPPED
created_at	2024-01-15

→ CDC detects UPDATE event

Only the changed record (status) is sent

3 Order is cancelled

Customer cancels the order

→ Production Database

orders	
id	5001
customer_id	101
amount	2500
status	CANCELLED
created_at	2024-01-15

→ CDC detects DELETE event

Delete event is sent (no need to move full data)

CDC Pipeline

Database (MySQL / PostgreSQL / Oracle etc.) → CDC Connector (Debezium / Kafka Connect / AWS DMS) → Apache Kafka (order-events topic) → Downstream Systems

Downstream Systems

- Data Warehouse (e.g., Snowflake, Redshift, BigQuery)
- Data Lake (e.g., S3, ADLS)
- Analytics / BI (e.g., Power BI, Tableau)
- Operational Systems (Inventory, Notifications, Fraud Detection)

Key Takeaways

- ✓ Capture only the changes (not the full table)
- ✓ Supports INSERT, UPDATE, DELETE
- ✓ Enables near real-time data pipelines
- ✓ Keeps all downstream systems in sync
- ✓ Common in e-commerce, banking, CRM, IoT, etc.

Tools Used for CDC

- Debezium (open source)
- Kafka Connect
- AWS DMS (managed)
- Google Datastream
- Azure Data Factory (CDC)
- Fivetran, Striim, etc.

Real-World Use Cases

- E-commerce order tracking
- Customer profile updates
- Inventory management
- Payment status updates
- Fraud detection
- Personalized recommendations

REAL-TIME DATA. REAL BUSINESS IMPACT.

Small Changes Big Possibilities!

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CDC - INTERVIEW CHEAT SHEET

Key Points, Tools, Advantages, Challenges & Interview Q&A

1. What is CDC?

- Change Data Capture (CDC) is a technique to capture only the changes (INSERT, UPDATE, DELETE) made to a database and send them to downstream systems in near real time.
- It reads database transaction logs and converts the changes into events.

2. CDC Captures

- INSERT → New records
- UPDATE → Modified records
- DELETE → Removed records

3. Common CDC Tools

- Debezium (open source)
- Kafka Connect
- AWS Database Migration Service (DMS)
- Google Datastream
- Azure Data Factory (CDC)
- Fivetran
- Striim
- Oracle GoldenGate

4. Advantages of CDC

- ✓ Near real-time data movement
- ✓ Move only changed data (reduces cost)
- ✓ Keeps systems in sync
- ✓ Supports event-driven architecture
- ✓ Scalable and efficient
- ✓ Useful for large databases
- ✓ Enables real-time analytics

5. Challenges of CDC

- ✗ Requires access to database logs
- ✗ Complex setup and configuration
- ✗ Handling schema changes
- ✗ Maintaining data consistency
- ✗ Dealing with large volumes of events
- ✗ Error handling and monitoring
- ✗ Tool-specific learning curve

6. Common Interview Questions

1. What is CDC and how does it work?
2. How is CDC different from incremental load?
3. How would you sync a large database with a data warehouse in near real time?
4. Which CDC tools have you used?
5. How do you handle schema changes in CDC?
6. What are the challenges in implementing CDC?

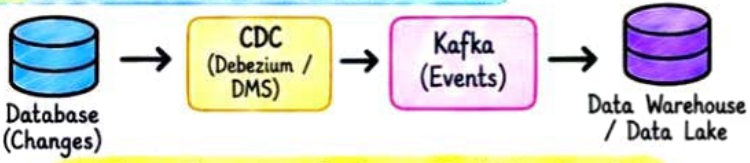
7. Sample Interview Answer

Q: How would you sync a production database to a data warehouse in near real time?

Answer: I would use a log-based CDC approach using tools like Debezium with Kafka Connect or a managed service like AWS DMS. The tool reads the database transaction logs, captures INSERT, UPDATE and DELETE events, publishes them to Kafka and the data is then consumed and loaded into the data warehouse (e.g., Snowflake/Redshift/BigQuery) in near real time. This ensures only changed data is moved, reducing cost and keeping both systems in sync.

Be Prepared
Be Confident
You Got This!

8. End-to-End Flow (Remember This!)



Database Changes → Events → Real-Time Insights

9. Final Takeaway

- ✓ Don't move everything.
- ✓ Move only what changed.
- ✓ CDC enables real-time data pipelines.
- ✓ It's a must-know topic for Data Engineers!

"Same data. Bigger possibilities with CDC!"